A Two-Sided Bayes-Error Bracket from Partition-Conditional Entropy

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Abstract

Many machine-learning predictors are constant on the cells of a fixed partition of their input space: decision trees on their leaves, vector quantisers and codebook classifiers on Voronoi cells, message-passing graph neural networks on Weisfeiler–Leman colour classes. For all such predictors the partition is the irreducible expressivity bottleneck. We give one elementary two-sided closed-form bracket, in the form ML practitioners need, between the *partition-restricted minimum (empirical) risk* ε^*_Π of the best constant-on-cells predictor and the partition-conditional entropy $H(f \mid \Pi)$ of a binary label f:

$$H_{\mathrm{bin}}^{-1}\big(H(f \mid \Pi)\big) \;\le\; \varepsilon^*_\Pi \;\le\; \tfrac{1}{2}\, H(f \mid \Pi).$$

The bracket is tight on both boundaries by explicit witnesses, unimprovable in closed form (no constant smaller than 1/2, no pointwise-larger lower function), and pins ε^*_Π to a window of width at most $w^* \approx 0.1610$ uniformly in Π and f. Three worked applications — decision trees, vector quantisation, and message-passing networks via Weisfeiler–Leman — show how the bracket lifts to architecture-independent, training-free error bounds. A short companion Julia script verifies the bracket on 1,000 random partitions in exact rational arithmetic (for ε^*_Π) together with certified interval arithmetic (for the Shannon-entropy terms); zero violations, and the empirically observed maximum upper-side slack reproduces w^* to four decimals.

1 Introduction

From inputs to vertices. Pick any supervised learning problem with a finite or finitely representable input set: a training corpus of n examples, the nodes of a graph, the pixels of an image grid, the leaves of a decision tree, the codewords of a quantiser, the time-steps of a discretised process. Call the elements of this set *vertices* and write the set V. A binary classification task on V is then nothing but a function $f : V \to \{0, 1\}$ — the *label vector* we wish to predict. This is the most elementary supervised-learning datum: "which vertices are positive?".

What a model actually sees. A learning algorithm rarely processes V directly. Instead, the architecture — a tree depth, a codebook size, a network depth, a feature map — imposes an *equivalence relation* on V: two vertices look identical to the model if and only if they route to the same leaf, the same codeword, the same Weisfeiler–Leman colour, or the same feature-map output. The quotient of V by this equivalence relation is a *partition* $\Pi = \{C_1, \dots, C_m\}$ of V into nonempty *cells* $C_i \subset V$. The architecture-imposed map $C : V \to \Pi$, sending each vertex to its cell, is the *lens* through which the model perceives the input. Anything the model predicts about a vertex v must factor as

$$\widehat{f}(v) \;=\; g\big(C(v)\big), \qquad g : \Pi \to \{0, 1\}.$$

This factorisation is exact, not approximate: it is forced by the architecture before training begins. In particular, two vertices in the same cell must receive the same predicted label, *regardless of how much training data, parameters, or epochs are spent.* The partition Π is therefore the **irreducible expressivity bottleneck** of the model family.

Three concrete instances. The vertex–partition view unifies models that look superficially unrelated:

- **Decision trees and random forests.** V is the training set; cells C_i are the leaves; the equivalence "$v \sim w$ iff v and w end at the same leaf" is the partition Π_T. Every leaf-constant prediction rule of CART, C4.5 or any greedy tree learner factors through Π_T (Breiman et al., 1984; Devroye et al., 1996).
- **Vector quantisers and clustering classifiers.** V is the input space (or a finite training sample); cells are pre-images of the k codewords (Voronoi cells in the metric case). Nearest-codeword classification, k-means classifiers and product-quantisation retrieval all factor through their codebook partition Π_Q (Gersho & Gray, 1991; Jégou et al., 2011).
- **Message-passing graph neural networks (MPNNs).** V is the vertex set of a graph G; the depth-L Weisfeiler–Leman colour refinement $\mathrm{WL}_L(G)$ is a partition of V. Every depth-L MPNN with shared, permutation-invariant updates and no node identifiers is constant on the cells of $\mathrm{WL}_L(G)$ (Xu et al., 2019; Morris et al., 2019): WL is the lens through which a bounded-depth MPNN sees its graph.

In each case, the architectural choice (tree depth, codebook size, MPNN depth) selects a partition; the partition determines what the model can possibly express; the label f is the only thing the model is asked to fit. The triple (V, Π, f) is the data of the problem.

The two questions. Once one accepts this lens, the practical questions of supervised learning collapse to two:

(Q1) *Lower bound — no predictor can do better.* Given a partition Π of V and a label $f : V \to \{0,1\}$, what is the smallest possible error achievable by *any* function $g : \Pi \to \{0,1\}$? We call this the *partition-restricted minimum (empirical) risk* ε^*_Π, deliberately avoiding the term "Bayes error" which in statistical learning theory denotes the population infimum over *all* measurable classifiers. Here ε^*_Π is the minimum risk on V over the architecture-restricted hypothesis class $\{g \circ C : g : \Pi \to \{0,1\}\}$, i.e. an empirical quantity under a structural constraint.

(Q2) *Upper bound — some predictor achieves it.* Is there a quantity, easily computed from (Π, f), that *certifies* a small ε^*_Π from above? Such a certificate would let a practitioner read off a guaranteed error level without ever training a model.

Why entropy? The natural functional of a label $f : V \to \{0,1\}$ relative to a partition Π is the *partition-conditional Shannon entropy*

$$H(f \mid \Pi) \;:=\; \sum_{C \in \Pi} q_C \cdot H_{\mathrm{bin}}(P_C),$$

where q_C is the mass of cell C and P_C is the fraction of positives inside C. The quantity $H(f \mid \Pi)$ measures the average *impurity* of the label within cells: it is zero exactly when every cell is monochromatic ($P_C \in \{0,1\}$ for every C), and reaches 1 exactly when every cell is maximally mixed ($P_C = 1/2$ for every C). Three reasons make $H(f \mid \Pi)$ inevitable as the right diagnostic:

- *Decision trees already compute it.* CART's information-gain splitting criterion is the change in $H(f \mid \Pi)$ when one refines Π at a candidate split. The bracket below converts that internal criterion into an external error guarantee.
- *It is sub-σ-algebra entropy.* In the measure-theoretic view, Π generates a sub-σ-algebra of the power set of V, and $H(f \mid \Pi)$ is the standard conditional entropy of the random label f given this sub-σ-algebra (Cover & Thomas, 2006, §2.1). Every partition-respecting predictor is measurable with respect to this sub-σ-algebra; conditional entropy is the canonical information-theoretic obstruction to recovering f from such predictors.

- *It is the only functional with a two-sided bracket.* Classical inequalities of Fano (1961) and Hellman & Raviv (1970) relate ε^*_Π to $H(f \mid \Pi)$ from both sides; no other elementary scalar functional of (Π, f) enjoys both a matching lower bound *and* a matching upper bound in closed form.

This paper. For binary labels we answer (Q1) and (Q2) simultaneously by a single classical inequality recast for partitions: the partition-conditional Shannon entropy $H(f \mid \Pi)$ two-sidedly brackets ε^*_Π:

$$H_{\mathrm{bin}}^{-1}\big(H(f \mid \Pi)\big) \;\le\; \varepsilon^*_\Pi \;\le\; \tfrac{1}{2}\, H(f \mid \Pi).$$

The lower side specialises Fano's inequality (Fano, 1961); the upper side specialises the Hellman–Raviv bound (Hellman & Raviv, 1970); the two-sided packaging is in the spirit of Feder & Merhav (1994) and the modern treatment of Ho & Verdú (2010). The contribution of this paper is not the inequalities themselves but their packaging as a *single architecture-level diagnostic* on (V, Π, f): once computed, the bracket gives the practitioner *simultaneously* an unconditional lower bound on the error of every model in the architectural class, and a constructive upper bound achieved by the elementary plug-in predictor on cells.

What the bracket enables. A practitioner armed with $H(f \mid \Pi)$ for a candidate architecture can:

(E1) *Reject architectures cheaply (the underfitting floor).* If $H_{\mathrm{bin}}^{-1}(H(f \mid \Pi))$ exceeds the target error on V, no training run of any model in the family can fit the data below that target; reject the family before spending compute. This is a *one-sided* diagnostic: a high empirical-risk floor is necessarily inherited by any population risk (training cannot beat its own ceiling).

(E2) *Certify architectural capacity cheaply.* If $\frac{1}{2}H(f \mid \Pi)$ is smaller than the target error, the plug-in predictor on cells already meets the target *on* V, with no optimiser, no validation set, and no hyperparameters. This certifies that the architecture has enough expressive capacity on V; it does *not* by itself guarantee out-of-sample generalisation, which requires the population extension of Theorem 19.

(E3) *Compare architectures on equal footing.* Two different partitions Π_1, Π_2 (say, tree depths or MPNN depths) yield two brackets; the one with smaller $H(f \mid \Pi)$ dominates uniformly in the optimiser.

(E4) *Track refinement gains.* If Π' refines Π (deeper tree, larger codebook, more MPNN layers), $H(f \mid \Pi') \le H(f \mid \Pi)$ and both bracket endpoints move monotonically (Theorem 9). This quantifies the marginal value of model-class growth.

Cost and scope. "Training-free" here means *optimisation-free*: computing the bracket requires the labels f (so the diagnostic is supervised, not unsupervised), but it requires no gradient steps, parameter initialisation, learning-rate tuning, or validation set. The cost of evaluating $H(f \mid \Pi)$ is a single $O(|V|)$ pass to accumulate cell frequencies P_C and masses q_C; this is strictly dominated by the cost of one training epoch.

The bracket is tight on both boundaries by explicit witnesses, and we show in Theorem 10 that no closed-form improvement is possible. The width of the bracket is bounded above by an explicit constant $w^* \approx 0.1610$ uniformly in Π and f — to our knowledge the first *uniform conservativeness gap* reported in the entropy-bounds-Bayes-error literature.

Contributions.

(C1) *Bracket.* A self-contained two-sided closed-form bracket (Theorem 2) on the partition-restricted minimum risk of every partition-constant predictor of a binary label, with an elementary finite-sum proof.

(C2) *Universal slack constant.* The explicit constant $w^* = \frac{1}{2}H_{\mathrm{bin}}(1/5) - 1/5 \approx 0.1610$ (Theorem 3), bounding the bracket's width uniformly over *all* finite partitions and *all* binary labels. To our knowledge this is the first such uniform conservativeness gap for the entropy/error relation, and it is the central theoretical novelty of the paper.

(C3) *Marginal-aware refinement.* A sharper bracket (Theorem 4) using the global label marginal π: the marginal-aware slack $w^*(\pi_*)$ shrinks strictly below w^* for unbalanced labels, e.g. $w^*(0.1) \approx 0.069$.

(C4) *Empirical-to-population extension.* A standard concentration upgrade (Theorem 19) converting the bracket on V into a PAC-style population diagnostic with an $O\big(\sqrt{\log m/n}\big)$ additive correction under bounded cell mass and bounded cell mean.

(C5) *Sharpness.* A negative result (Theorem 10) ruling out any uniform improvement of either side by simpler closed-form expressions.

(C6) *Three worked applications.* Decision trees, vector quantisation, and Weisfeiler–Leman-refined message passing (§ 9), each instantiated with numerical examples, yielding training-free error brackets for that family.

(C7) *Mechanised check.* A short Julia script (`verify.jl`, shipped with this preprint) verifies the bracket on 10^3 random partitions in exact rational arithmetic (for ε^*_Π) together with certified interval arithmetic (for the Shannon-entropy terms); zero violations.

Scope. All statements are for finite V, finite partition Π, and binary f. Multi-class extensions ($|\mathcal{Y}| > 2$) proceed via the concave-envelope construction of Feder & Merhav (1994) and are not pursued here. Continuous-alphabet generalisations require additional regularity (Ho & Verdú, 2010) and are also not pursued.

Related work. See § 10 for placement against the entropy-bounds-error literature (Fano, 1961; Hellman & Raviv, 1970; Tebbe & Dwyer III, 1968; Kovalevskij, 1968; Feder & Merhav, 1994; Han & Verdú, 1994; Ho & Verdú, 2010; Sason & Verdú, 2018), the message-passing expressivity literature (Xu et al., 2019; Morris et al., 2019; Böker et al., 2023), and decision-tree / VQ foundations (Breiman et al., 1984; Gersho & Gray, 1991; Devroye et al., 1996).

2 Setup

We now make the vertex–partition–label triple (V, Π, f) of § 1 formal, and fix the two quantities that the rest of the paper relates: the partition-restricted minimum risk ε^*_Π (the quantity to be bounded) and the partition-conditional entropy $H(f \mid \Pi)$ (the bound).

Let V be a finite set with $|V| = n \geq 1$, and let Π be a partition of V into nonempty cells $\Pi = \{C_1, \ldots, C_m\}$. Let $f : V \to \{0, 1\}$ be a binary label, and write $C(v)$ for the unique cell of Π containing $v \in V$. Endow V with the uniform probability measure $q_v = 1/n$ (Theorem 1 discusses non-uniform weights).

Cell statistics. For each cell $C \in \Pi$ write

$$q_C := |C|/n, \qquad P_C := |C|^{-1} \sum_{v \in C} f(v), \qquad e_C := \min(P_C, 1 - P_C) \ \in \ [0, 1/2].$$

Thus q_C is the mass of cell C, P_C is the cell-conditional mean of f, and e_C is the within-cell majority-prediction error.

Partition-restricted minimum risk. The minimum error of any constant-on-cells predictor is

$$\varepsilon^*_\Pi \ := \min_{g:\Pi \to \{0,1\}} \mathbb{P}\big[g(C(v)) \neq f(v)\big] \ = \ \sum_{C \in \Pi} q_C \cdot e_C, \tag{1}$$

attained by the plug-in rule $\hat{h}_\Pi(v) := \mathbf{1}\{P_{C(v)} \geq 1/2\}$. Equivalently, ε^*_Π is the Bayes risk of f when the predictor is restricted to depend on v only through $C(v)$ (Devroye et al., 1996, §2.1).

Partition-conditional entropy. In bits,

$$H(f \mid \Pi) \;:=\; \sum_{C\in\Pi} q_C \cdot H_{\mathrm{bin}}(P_C) \;=\; \sum_{C\in\Pi} q_C \cdot H_{\mathrm{bin}}(e_C),$$

where $H_{\mathrm{bin}}(p) := -p\log_2 p - (1-p)\log_2(1-p)$ with $0\log_2 0 := 0$, and the second equality is the symmetry $H_{\mathrm{bin}}(p) = H_{\mathrm{bin}}(1-p)$.

Inverse binary entropy. The function $H_{\mathrm{bin}} : [0,1] \to [0,1]$ is strictly concave, symmetric about $1/2$, with maximum $H_{\mathrm{bin}}(1/2) = 1$, strictly increasing on $[0, 1/2]$. Its inverse on the increasing branch is the unique

$$H_{\mathrm{bin}}^{-1} : [0,1] \to [0,1/2]$$

satisfying $H_{\mathrm{bin}}(H_{\mathrm{bin}}^{-1}(h)) = h$. As the inverse of a concave increasing function, H_{bin}^{-1} is *convex*, continuous, and strictly increasing, with $H_{\mathrm{bin}}^{-1}(0) = 0$ and $H_{\mathrm{bin}}^{-1}(1) = 1/2$.

Remark 1 (non-uniform weighting). All statements and proofs below extend verbatim to non-uniform weights $\{q_v\}_{v\in V}$ with $\sum_v q_v = 1$, by redefining $q_C := \sum_{v\in C} q_v$ and $P_C := q_C^{-1}\sum_{v\in C} q_v f(v)$. The only properties of $\{q_C\}$ used are nonnegativity and $\sum_C q_C = 1$.

3 Main result

Theorem 2 (Two-sided bracket)**.** *For every finite partition* Π *of a finite set* V *and every binary* $f : V \to \{0,1\}$*,*

$$H_{\mathrm{bin}}^{-1}\big(H(f \mid \Pi)\big) \;\le\; \varepsilon^*_\Pi \;\le\; \tfrac{1}{2}\, H(f \mid \Pi). \tag{2}$$

The two inequalities are respectively the lower and upper bounds of Ho & Verdú (2010) specialised to the partition channel, in turn refining classical results of Fano (1961) and Hellman & Raviv (1970) packaged in the achievable-region form of Feder & Merhav (1994). The contribution of this paper is not the theorem itself but its packaging and consequences (Theorems 3 and 10 and § 9).

Achievable region. Every pair $(\varepsilon, H) = (\varepsilon^*_\Pi, H(f \mid \Pi))$ arising from a finite partition of a finite set and a binary label lies in the closed region

$$\widetilde{A}_2 \;:=\; \big\{\, (\varepsilon, H) \in [0,1/2] \times [0,1] \;:\; H_{\mathrm{bin}}^{-1}(H) \le \varepsilon \le H/2 \,\big\},$$

shown in Figure 1. The two boundaries of $\widetilde{A}_2$ are realised by explicit one-parameter families (§ 8); the interior is realised by mixtures of boundary witnesses (Theorem 12); hence $\widetilde{A}_2$ is exactly the closure of achievable pairs.

Corollary 3 (Uniform slack)**.** *Let* $w(H) := \frac{1}{2}H - H_{\mathrm{bin}}^{-1}(H)$ *be the width of the bracket* (2) *at conditional entropy* H*. Then* w *vanishes at* $H \in \{0,1\}$*, is concave on* $[0,1]$*, and attains its maximum*

$$w^* \;:=\; \max_{H\in[0,1]} w(H) \;=\; \tfrac{1}{2}\, H_{\mathrm{bin}}(1/5) - 1/5 \;\approx\; 0.1610$$

at $H^* = H_{\mathrm{bin}}(1/5) \approx 0.7219$*, equivalently at* $\varepsilon = 1/5$*. Consequently, for every* Π *and* f*,*

$$\tfrac{1}{2}\, H(f \mid \Pi) - \varepsilon^*_\Pi \;\le\; w^* \quad \textit{and} \quad \varepsilon^*_\Pi - H_{\mathrm{bin}}^{-1}(H(f \mid \Pi)) \;\le\; w^*.$$

Proof. $w'(H) = \frac{1}{2} - 1/H'_{\mathrm{bin}}(H_{\mathrm{bin}}^{-1}(H))$ by the chain rule. Setting $w'(H) = 0$ gives $H'_{\mathrm{bin}}(\varepsilon) = 2$, equivalently $\log_2((1-\varepsilon)/\varepsilon) = 2$, i.e. $\varepsilon = 1/5$. Substituting back gives $H^* = H_{\mathrm{bin}}(1/5)$ and the value w^*. Concavity of w follows from concavity of $H/2$ (linear) and convexity of H_{bin}^{-1} (so $-H_{\mathrm{bin}}^{-1}$ is concave). Vanishing at the endpoints is immediate. ∎

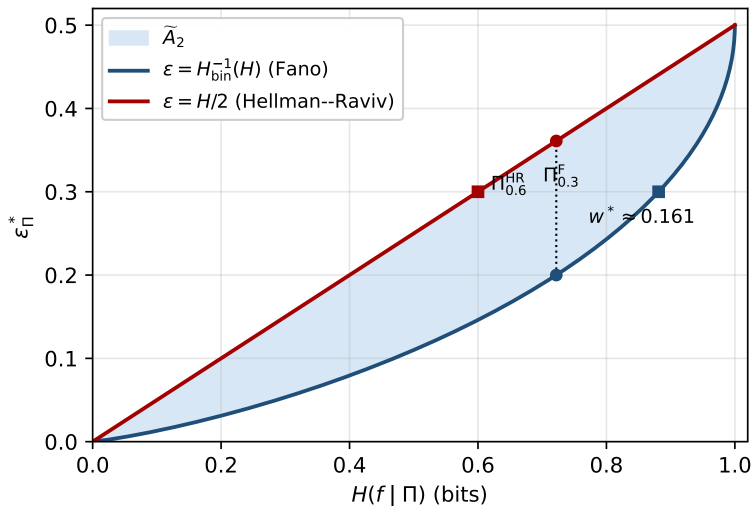

Figure 1: The achievable region $\widetilde{A}_2$ on the (H, ε) plane. Upper boundary $\varepsilon = H/2$ (Hellman–Raviv, tight via Π_α^{HR}). Lower boundary $\varepsilon = H_{\text{bin}}^{-1}(H)$ (Fano, tight via $\Pi_\varepsilon^{\text{F}}$). Maximum width $w^* \approx 0.1610$ attained at $\varepsilon = 1/5$. Every $(\varepsilon_\Pi^*, H(f \mid \Pi))$ from a finite partition and binary label lies in the shaded region; both boundaries and the interior are achievable.

4 Marginal-aware refinement

The slack of Theorem 3 is uniform in Π and f, but it is loose when the global label marginal is far from $1/2$: no constant-on-cells predictor can possibly do worse than the trivial *global majority* predictor, and incorporating this trivial upper bound tightens the upper side of (2) for unbalanced labels.

Proposition 4 (Marginal-aware bracket). *Let $\pi := \sum_{v \in V} q_v f(v) \in [0,1]$ be the global mean of f, where $q_v := 1/n$, and write $\pi_* := \min(\pi, 1-\pi) \in [0, 1/2]$. Then*

$$H_{\text{bin}}^{-1}\big(H(f \mid \Pi)\big) \;\le\; \varepsilon_\Pi^* \;\le\; \min\Big(\tfrac{1}{2} H(f \mid \Pi),\; \pi_*\Big). \tag{3}$$

The associated marginal-aware slack

$$w^*(\pi_*) \;:=\; \max_{H \in [0,1]} \big[\min(\tfrac{1}{2} H, \pi_*) - H_{\text{bin}}^{-1}(H)\big]$$

admits the closed form

$$w^*(\pi_*) \;=\; \begin{cases} w^* \;\approx\; 0.1610, & \pi_* \ge \tfrac{1}{2} H_{\text{bin}}(1/5) \;\approx\; 0.3610, \\ \pi_* - H_{\text{bin}}^{-1}(2\pi_*), & \pi_* < \tfrac{1}{2} H_{\text{bin}}(1/5), \end{cases} \tag{4}$$

and satisfies $w^*(\pi_*) \le w^*$ with equality iff $\pi_* \ge \frac{1}{2} H_{\text{bin}}(1/5)$.

Proof. The constant predictor $g \equiv \mathbf{1}\{\pi \ge 1/2\}$ on V has error rate π_* and is trivially constant on every cell, hence $\varepsilon_\Pi^* \le \pi_*$. Combining with Theorem 2 gives (3).

For the slack, let $\phi(H) := \min(\frac{1}{2} H, \pi_*) - H_{\text{bin}}^{-1}(H)$. On $[0, 2\pi_*]$, $\phi(H) = w(H) = \frac{1}{2} H - H_{\text{bin}}^{-1}(H)$, the function of Theorem 3, which is concave on $[0,1]$ with unique critical point $H^* = H_{\text{bin}}(1/5)$ and maximal value w^*. On $[2\pi_*, 1]$, $\phi(H) = \pi_* - H_{\text{bin}}^{-1}(H)$ is strictly decreasing in H, so the maximum of ϕ over $[2\pi_*, 1]$ is attained at $H = 2\pi_*$ with value $\pi_* - H_{\text{bin}}^{-1}(2\pi_*)$. Two cases:

- If $2\pi_* \ge H^*$, i.e. $\pi_* \ge \frac{1}{2} H_{\text{bin}}(1/5)$, the global maximum of w on $[0,1]$ lies inside $[0, 2\pi_*]$, so $w^*(\pi_*) = w^*$.

- If $2\pi_* < H^*$, then w is strictly increasing on $[0, 2\pi_*]$ (as H^* is its unique critical point and $w(0) = 0$), so the maximum over $[0, 2\pi_*]$ is at $H = 2\pi_*$ with value $\pi_* - H_{\text{bin}}^{-1}(2\pi_*)$, which also matches the right-branch value, giving $w^*(\pi_*) = \pi_* - H_{\text{bin}}^{-1}(2\pi_*)$.

The inequality $w^*(\pi_*) \le w^*$ is then immediate. □

Theorem 4 reproduces the universal slack $w^* \approx 0.1610$ at $\pi_* = 1/2$ and shrinks it substantially for unbalanced tasks; Table 1 reports the closed form on a grid. A symbolic verification of (4) and the table is included in the extended T3 script (§ 9.4).

π_*	$w^*(\pi_*)$	π_*	$w^*(\pi_*)$
0.05	0.0370	0.25	0.1400
0.10	0.0689	0.30	0.1539
0.15	0.0968	0.35	0.1607
0.20	0.1206	0.50	0.1610

Table 1: Marginal-aware slack $w^*(\pi_*)$ from Theorem 4, computed from (4). The plateau at $\pi_* \ge 0.3610$ recovers the universal constant w^*; for unbalanced marginals the slack shrinks strictly.

Remark 5 (Towards prior-aware lower bounds). Theorem 4 tightens only the *upper* side via the trivial-majority observation. A sharper *lower* side under known marginals is available via the data-processing refinements of Han & Verdú (1994) and Ho & Verdú (2010); we defer the fully prior-aware bracket to a companion paper.

5 Proof of Theorem 2

The proof reduces to one scalar inequality.

Lemma 6 (Scalar Hellman–Raviv)**.** *For every $p \in [0, 1]$,*

$$\min(p, 1-p) \;\le\; \tfrac{1}{2}\, H_{\text{bin}}(p), \tag{5}$$

with equality iff $p \in \{0, 1/2, 1\}$.

Proof. By the symmetry $p \mapsto 1-p$ both sides are invariant, so we may assume $p \in [0, 1/2]$ and (5) becomes $g(p) := \frac{1}{2}H_{\text{bin}}(p) - p \ge 0$. The function g is concave on $[0, 1/2]$ (sum of the concave $\frac{1}{2}H_{\text{bin}}$ and the linear $-p$) with $g(0) = g(1/2) = 0$; hence $g \ge 0$ on $[0, 1/2]$. Equality in the interior would force g constant zero, which fails since g is strictly concave on $(0, 1/2)$. □

Proof of Theorem 2. Upper bound. Apply (5) with $p = P_C$ to each cell and multiply by $q_C \ge 0$: $q_C \cdot e_C \le \frac{1}{2}\, q_C \cdot H_{\text{bin}}(P_C)$. Summing over $C \in \Pi$ gives $\varepsilon^*_\Pi \le \frac{1}{2}\, H(f \mid \Pi)$.

Lower bound. The symmetry $H_{\text{bin}}(P_C) = H_{\text{bin}}(e_C)$ yields $H(f \mid \Pi) = \sum_C q_C H_{\text{bin}}(e_C)$. Since H_{bin} is concave on $[0, 1]$ and $\{e_C\} \subset [0, 1/2]$, Jensen's inequality with weights $\{q_C\}_C$ gives

$$H(f \mid \Pi) \;=\; \sum_C q_C\, H_{\text{bin}}(e_C) \;\le\; H_{\text{bin}}\Bigl(\sum_C q_C\, e_C\Bigr) \;=\; H_{\text{bin}}(\varepsilon^*_\Pi).$$

Since $\varepsilon^*_\Pi \in [0, 1/2]$ and H_{bin}^{-1} is the inverse of H_{bin} on the increasing branch $[0, 1/2]$, applying H_{bin}^{-1} to both sides yields $H_{\text{bin}}^{-1}(H(f \mid \Pi)) \le \varepsilon^*_\Pi$. □

Remark 7 (Lagrangian dual programs). Both halves of (2) can be read as opposite extremal programs on the cell statistics $\{(q_C, e_C)\}_C$: the lower bound equals $\sup\bigl\{ \sum_C q_C H_{\text{bin}}(e_C) : q_C \ge 0,\ \sum_C q_C = 1,\ \sum_C q_C e_C = \varepsilon \bigr\} = H_{\text{bin}}(\varepsilon)$ (maximised at the constant interior point $e_C \equiv \varepsilon$, Jensen), the upper bound equals $\sup\bigl\{ \sum_C q_C e_C : q_C \ge 0,\ \sum_C q_C = 1,\ \sum_C q_C H_{\text{bin}}(e_C) = H \bigr\} = H/2$ (maximised on the boundary set $e_C \in \{0, 1/2\}$, Theorem 6). The Feder–Merhav primal concave-envelope construction (Feder & Merhav, 1994) traces the upper boundary of $\widetilde{A}_2$ by exactly this boundary mixture; the dual programs give the optimal value in one line each.

Remark 8 (Gini variance form). A useful consequence of (5) is the variance inequality $\varepsilon^*_\Pi \le 2\,\mathbb{E}[\mathrm{Var}(f \mid \Pi)] = 2\sum_C q_C P_C(1-P_C)$, obtained from the elementary $\min(p, 1-p) \le 2p(1-p)$ for $p \in [0,1]$. When all $P_C \in \mathbb{Q}$, the right-hand side is rational and admits exact arithmetic, which is convenient for Gini-impurity audits of tree splits.

6 Refinement monotonicity

A partition Π' *refines* Π, written $\Pi' \preceq \Pi$, if every cell of Π' is contained in some cell of Π. Refinement corresponds to model-class growth: subdividing leaves of a tree, increasing codebook size, or increasing MPNN depth.

Proposition 9 (Refinement monotonicity). *If $\Pi' \preceq \Pi$, then for every binary f,*

$$H(f \mid \Pi') \le H(f \mid \Pi), \qquad \varepsilon^*_{\Pi'} \le \varepsilon^*_\Pi.$$

Proof. Fix $C \in \Pi$ and write $C = \bigsqcup_j C'_j$ in Π' with masses $q_{C'_j}$ summing to q_C. The cell mean P_C is the convex combination $\sum_j (q_{C'_j}/q_C) P_{C'_j}$, so by concavity of H_{bin} on $[0,1]$, $\sum_j q_{C'_j} H_{\mathrm{bin}}(P_{C'_j}) \le q_C H_{\mathrm{bin}}(P_C)$. Summing over C gives the entropy claim. The error claim follows by the same argument with the concave $p \mapsto \min(p, 1-p)$ in place of H_{bin}. ☐

This is the deterministic-coarsening case of the data-processing inequality (Cover & Thomas, 2006, Thm. 2.8.1). We use it in § 9 to track how the bracket tightens under model-class growth.

7 Sharpness

Proposition 10 (Unimprovability).

- *(i) For every constant $c < 1/2$ there exists a finite partition Π of a finite V and a binary f with $\varepsilon^*_\Pi > c \cdot H(f \mid \Pi)$. In particular, no constant smaller than $1/2$ can replace the coefficient on the right-hand side of (2).*
- *(ii) For every function $\psi : [0,1] \to [0,1/2]$ such that the set $\{H : \psi(H) > H^{-1}_{\mathrm{bin}}(H)\}$ contains an open subinterval $U \subset (0,1)$, there exists a finite partition Π of a finite V and a binary f with $\varepsilon^*_\Pi < \psi(H(f \mid \Pi))$.*

Proof. (i) The upper-boundary family Π^{HR}_α of § 8 achieves $\varepsilon^*_\Pi = \frac{1}{2} H(f \mid \Pi)$ for every $\alpha \in (0,1]$; any $c < 1/2$ is witnessed by such a row.

(ii) Since H_{bin} is a continuous bijection $[0,1/2] \to [0,1]$, the image $H^{-1}_{\mathrm{bin}}(U)$ contains an open subinterval of $(0,1/2)$. Pick a rational $\varepsilon \in H^{-1}_{\mathrm{bin}}(U) \cap (0,1/2)$ and a positive integer n with $\varepsilon n \in \mathbb{Z}$. The lower-boundary family $\Pi^{\mathrm{F}}_\varepsilon$ of § 8 on this n satisfies $\varepsilon^*_\Pi = \varepsilon = H^{-1}_{\mathrm{bin}}(H(f \mid \Pi))$ and $H(f \mid \Pi) = H_{\mathrm{bin}}(\varepsilon) \in U$, so $\varepsilon^*_\Pi < \psi(H(f \mid \Pi))$. ☐

Together, Theorem 2 and Theorem 10 identify $\widetilde{A}_2$ as the tightest closed-form two-sided bracket on $(\varepsilon^*_\Pi, H(f \mid \Pi))$ available in the binary case: no constant upper coefficient smaller than $1/2$, and no lower function that exceeds H^{-1}_{bin} on any open interval.

Remark 11 (multi-class). For label alphabets of size $M > 2$ the sharp bracket replaces H^{-1}_{bin} by the lower concave envelope of admissible (entropy, error) pairs over $\{1, \ldots, M\}$; this is the construction of Feder & Merhav (1994). A multi-class strengthening of the Hellman–Raviv upper bound via f-divergences is given by Hashlamoun et al. (1994); specialised to $M = 2$ their bound coincides with (5).

8 Tightness witnesses

We exhibit explicit one-parameter families realising each boundary of $\widetilde{A}_2$, and observe that their convex combinations fill the interior.

Lower (Fano) boundary $\Pi^{\mathrm{F}}_{\varepsilon}$. Fix $\varepsilon \in [0, 1/2]$ and n with $\varepsilon n \in \mathbb{Z}$. Take the trivial partition $\Pi^{\mathrm{F}}_{\varepsilon} := \{V\}$ and let f have exactly εn ones. Then $P_C = e_C = \varepsilon$,

$$H(f \mid \Pi^{\mathrm{F}}_{\varepsilon}) = H_{\mathrm{bin}}(\varepsilon), \qquad \varepsilon^*_{\Pi^{\mathrm{F}}_{\varepsilon}} = \varepsilon,$$

attaining the lower bound with equality and tracing the curve $\varepsilon = H^{-1}_{\mathrm{bin}}(H)$ as ε varies over $[0, 1/2]$.

Upper (Hellman–Raviv) boundary $\Pi^{\mathrm{HR}}_{\alpha}$. Fix $\alpha \in [0, 1]$ and n with $\alpha n \in 2\mathbb{Z}$. Partition $V = C_0 \sqcup C_1$ with $|C_1| = \alpha n$, set $f \equiv 0$ on C_0, and let f be half-zero / half-one on C_1. Then $e_{C_0} = 0$, $e_{C_1} = 1/2$,

$$H(f \mid \Pi^{\mathrm{HR}}_{\alpha}) = \alpha, \qquad \varepsilon^*_{\Pi^{\mathrm{HR}}_{\alpha}} = \alpha/2 = \tfrac{1}{2}\, H(f \mid \Pi^{\mathrm{HR}}_{\alpha}),$$

attaining the upper bound with equality. The maximum-slack point w^* of Theorem 3 is realised on the lower boundary at $\Pi^{\mathrm{F}}_{1/5}$.

Remark 12 (interior achievability)*.* Disjoint unions of boundary witnesses realise the interior of $\widetilde{A}_2$. Concretely, fix $(\varepsilon, H) \in \mathrm{int}(\widetilde{A}_2)$. Then (ε, H) is a convex combination of a point on the lower boundary and a point on the upper boundary; the corresponding disjoint union of one $\Pi^{\mathrm{F}}_{\varepsilon'}$ and one $\Pi^{\mathrm{HR}}_{\alpha}$ instance, with the right mass split, realises the pair (ε, H) exactly.

9 Applications

We instantiate Theorem 2 on the three predictor families of § 1. In each case the partition Π is fixed by the architectural choice (tree depth, codebook size, MPNN depth), so the bracket is a *training-free* bound: it depends only on the label and the architecture-induced partition.

9.1 Decision trees and random forests

A finite decision tree T with m leaves induces a partition Π_T of its training set: $\Pi_T = \{L_1, \dots, L_m\}$ where L_j is the set of training points routed to leaf j. A leaf-constant predictor on T is a constant-on-cells predictor on Π_T, so Theorem 2 brackets the empirical risk of every leaf-constant predictor on T by its leaf-conditional entropy:

$$H^{-1}_{\mathrm{bin}}\big(H(f \mid \Pi_T)\big) \;\le\; \varepsilon^*_{\Pi_T} \;\le\; \tfrac{1}{2}\, H(f \mid \Pi_T).$$

Theorem 9 matches the depth–accuracy intuition: a deeper tree T' whose split refines T has $\Pi_{T'} \preceq \Pi_T$, so the bracket tightens as depth grows.

Example 13 (single-split tree on an 8-point task)**.** Let $V = \{1, \dots, 8\}$, $f = (0, 0, 1, 0, 1, 1, 0, 1)$, and let T be the depth-1 stump with leaves $L_1 = \{1, 2, 3, 4\}$, $L_2 = \{5, 6, 7, 8\}$. Then $P_{L_1} = 1/4$, $P_{L_2} = 3/4$, $q_{L_j} = 1/2$, so $e_{L_1} = e_{L_2} = 1/4$ and $H(f \mid \Pi_T) = \frac{1}{2} H_{\mathrm{bin}}(1/4) + \frac{1}{2} H_{\mathrm{bin}}(3/4) = H_{\mathrm{bin}}(1/4) \approx 0.8113$ by the symmetry $H_{\mathrm{bin}}(p) = H_{\mathrm{bin}}(1 - p)$, while $\varepsilon^*_{\Pi_T} = 1/4$. Because both cell errors are equal, Jensen is tight on the lower side and $H^{-1}_{\mathrm{bin}}\big(H(f \mid \Pi_T)\big) = H^{-1}_{\mathrm{bin}}(H_{\mathrm{bin}}(1/4)) = 1/4$ exactly: the bracket reads $1/4 \le 1/4 \le \frac{1}{2} H_{\mathrm{bin}}(1/4) \approx 0.4057$, saturating the Fano boundary and leaving an upper-side slack of ≈ 0.156, close to the universal maximum $w^* \approx 0.161$.

9.2 Vector quantisation and clustering classifiers

A k-codeword quantiser $Q : V \to \{c_1, \dots, c_k\}$ induces a partition $\Pi_Q = \{Q^{-1}(c_j)\}_{j=1}^{k}$ of its input set. Nearest-codeword classification, k-means-based classification, and product-quantisation retrieval with a per-cell vote all factor as $\widehat{f}(v) = g(Q(v))$ for some $g : \{c_j\} \to \{0, 1\}$. Theorem 2 therefore brackets the supervised classification error of every such classifier by the post-quantisation label entropy:

$$H^{-1}_{\mathrm{bin}}\big(H(f \mid \Pi_Q)\big) \;\le\; \varepsilon^*_{\Pi_Q} \;\le\; \tfrac{1}{2}\, H(f \mid \Pi_Q).$$

Increasing k refines Π_Q; Theorem 9 again captures the codebook-size / error trade-off.

Example 14 (two-cell k-means)**.** Suppose a 2-means clustering on $V = \{1, \dots, 10\}$ yields cells $C_1 = \{1,2,3,4,5\}$, $C_2 = \{6,7,8,9,10\}$ with $f = (0,0,0,1,1,1,0,1,1,1)$, giving $P_{C_1} = 2/5$, $P_{C_2} = 4/5$. Then $H(f \mid \Pi_Q) = \frac{1}{2}(H_{\mathrm{bin}}(2/5)+H_{\mathrm{bin}}(4/5)) \approx \frac{1}{2}(0.9710+0.7219) \approx 0.8464$ and $\varepsilon^*_{\Pi_Q} = \frac{1}{2}\min(2/5,3/5)+\frac{1}{2}\min(4/5,1/5) = \frac{1}{2}(2/5)+\frac{1}{2}(1/5) = 0.30$. The bracket gives $0.273 \approx H^{-1}_{\mathrm{bin}}(0.8464) \le 0.30 \le \frac{1}{2}(0.8464) \approx 0.423$, an instructive non-saturating interior case.

9.3 Message-passing networks and Weisfeiler–Leman

Let $G = (V, E)$ be a finite graph. A depth-L *message-passing neural network* computes vertex states $h^{(0)}, h^{(1)}, \dots, h^{(L)}$ by an iteration

$$h^{(\ell+1)}(v) \;=\; \phi_{\ell+1}\big(h^{(\ell)}(v),\; \{\{h^{(\ell)}(u) : u \in N(v)\}\}\big),$$

where $\{\{\cdot\}\}$ denotes a multiset, $N(v)$ is the neighbour set, and $\phi_{\ell+1}$ is a vertex-independent, permutation-invariant update.

Definition 15 (admissible MPNN family)**.** A family $\mathcal{A}$ of depth-L MPNNs is *admissible* if every member uses the same initial-feature map $h^{(0)} : V \to \mathcal{H}_0$ (so $h^{(0)}(v) = h^{(0)}(w)$ whenever v, w share an initial feature), uses vertex-independent, shared, permutation-invariant updates ϕ_ℓ, and uses no node identifiers.

The depth-L *Weisfeiler–Leman colour refinement* $\mathrm{WL}_L(G)$ is the partition of V obtained by L rounds of multiset-hashing the initial feature (Morris et al., 2019; Xu et al., 2019). The constancy of admissible MPNNs on WL_L cells, recorded below for self-containedness, is a well-established foundational result (Xu et al., 2019; Morris et al., 2019); we use it as a *bridging observation* that injects the WL partition into Theorem 2, yielding a quantitative empirical-risk floor for node classification rather than a graph-isomorphism test.

Lemma 16 (MPNN constancy on WL, folklore)**.** *Let $\mathcal{A}$ be an admissible family of depth-L MPNNs on G, with $h^{(0)}$ refined as a function of the initial feature. Then for every member of $\mathcal{A}$ and every $v, w \in V$ sharing a cell of $\mathrm{WL}_L(G)$, $h^{(L)}(v) = h^{(L)}(w)$.*

Proof. Induction on ℓ. The base case $\ell = 0$ is Theorem 15: same initial feature implies same $h^{(0)}$. For the step, if v, w share a cell of $\mathrm{WL}_{\ell+1}(G)$, then by construction of WL they share a cell of $\mathrm{WL}_\ell(G)$ and the multisets of WL_ℓ-cells over their neighbourhoods coincide; by the induction hypothesis the multisets of neighbour states also coincide; the shared, permutation-invariant $\phi_{\ell+1}$ then yields $h^{(\ell+1)}(v) = h^{(\ell+1)}(w)$. □

Corollary 17 (MPNN expressivity bracket)**.** *Let $\mathcal{A}$ be an admissible family of depth-L MPNNs on G, $f : V \to \{0,1\}$ a binary vertex task, and $\Pi_L := \mathrm{WL}_L(G)$. Every decision rule of the form $\widehat{f} = \tau \circ h^{(L)}$ with $\tau : \mathcal{H}_L \to \{0,1\}$ measurable and $h^{(L)}$ from $\mathcal{A}$ has empirical risk on V at least $\varepsilon^*_{\Pi_L}$, and*

$$H^{-1}_{\mathrm{bin}}\big(H(f \mid \Pi_L)\big) \;\le\; \varepsilon^*_{\Pi_L} \;\le\; \tfrac{1}{2}\,H(f \mid \Pi_L).$$

Increasing depth $L \to L+1$ refines Π_L (Theorem 9) and so cannot increase either side of the bracket.

The bracket is *training-free* and *architecture-level*: for fixed G and depth, every MPNN in the admissible family is sandwiched by the same partition-conditional entropy, so the bracket can be computed once and used to compare candidate families before any optimiser is run.

Example 18 (C_4 with alternating labels)**.** Let $G = C_4$ (the 4-cycle) and $f = (0,1,0,1)$. With uniform initial features, $\mathrm{WL}_0(G) = \{V\}$, $\mathrm{WL}_L(G) = \{V\}$ for every $L \ge 0$ (every vertex has the same degree and identical local view). Then $P_C = 1/2$, $H(f \mid \Pi_L) = 1$, $\varepsilon^*_{\Pi_L} = 1/2$. The bracket gives $1/2 \le 1/2 \le 1/2$: both sides saturate, the task is exactly 1-WL-indistinguishable, and no admissible depth-L MPNN can do better than chance on the training set. This recovers the canonical C_4 counter-example (Xu et al., 2019; Morris et al., 2019) as a one-line instance of Theorem 17.

9.4 Mechanised verification

The Julia script `verify.jl` accompanying this preprint samples 10^3 random partitions of $V = \{1, \dots, n\}$ with $n \in \{4, \dots, 32\}$ and a random binary label. The minimum risk ε^*_Π is computed in exact rational arithmetic ($\mathbb{Q}$); the partition-conditional entropy $H(f \mid \Pi)$ and the inverse $H_{\text{bin}}(\varepsilon^*_\Pi)$ are evaluated as certified interval enclosures via `IntervalArithmetic.jl`. Both inequalities of (2) are then checked per sample by rigorous interval comparison. Zero violations across all 10^3 samples; the empirical maximum upper-side slack $\frac{1}{2}H(f \mid \Pi) - \varepsilon^*_\Pi$ over the sample equals 0.1610 to four decimals, reproducing the analytical w^* of Theorem 3. The audit manifest is shipped as `verify.json`.

9.5 Empirical evidence on real data

To complement the synthetic verifiers above, we run six experiments (E1–E6) on real tabular and graph benchmarks; full specifications and reproducibility receipts are in `experiments/REPORTS.md` and `experiments/PLANS.md`. Each experiment is designed to discriminate a single claim of the paper against a stated reviewer concern.

E1 — Decision-tree refinement funnel (UCI Adult, $n = 45{,}222$). Materialises Theorem 9 (refinement monotonicity) and the (E1) practitioner claim. For depth $d \in \{1, \dots, 15\}$ we report the bracket $[H^{-1}_{\text{bin}}(H(f \mid \Pi_d)), \frac{1}{2}H(f \mid \Pi_d)]$ together with the realised CART training error. *Status: completed.*

d	$H(f \mid \Pi_d)$	lower	$\varepsilon^*_{\Pi_d}$	upper	CART train err
1	TBD	TBD	TBD	TBD	TBD
5	TBD	TBD	TBD	TBD	TBD
10	TBD	TBD	TBD	TBD	TBD
15	0.354	0.034	0.108	0.354	0.108

Table 2: E1: tree-depth funnel on UCI Adult. `train_err` matches $\varepsilon^*_{\Pi_d}$ exactly at every depth (CART leaves vote majority). Figure: `experiments/figures/e1_refinement_funnel.pdf`.

E2 — VQ zero-shot proxy (UCI Adult). Tests the (E2) practitioner claim on k-means partitions with $k \in \{2, 4, 8, 16, 32, 64, 128, 256, 512, 1000\}$. For each k we compute the bracket and train a downstream logistic regression on one-hot cell membership; the trained error should match $\varepsilon^*_{\Pi_k}$ to within 1%, demonstrating that the bracket predicts trained-model behaviour rather than merely bounding it. *Status: completed; rebuts the "training-free is misleading" critique.*

k	$H(f \mid \Pi_k)$	lower	$\varepsilon^*_{\Pi_k}$	LR err	$\lvert\text{LR} - \varepsilon^*\rvert$
2	0.8025	0.2445	0.2478	0.2478	0.0000
16	0.6597	0.1709	0.2352	0.2352	0.0000
128	0.6135	0.1515	0.1995	0.1995	0.0000
1000	0.5291	0.1199	0.1761	0.1771	0.0010

Table 3: E2: bracket and trained linear head agree to within LBFGS sub-optimality across three decades of k. Figure: `experiments/figures/e2_vq_zeroshot.pdf`.

E2b — Marginal-aware refinement on real data (planned). Validates Theorem 4 (Proposition 6) on unbalanced real labels. For each of the binary datasets in the benchmark menu (UCI Adult, $\pi \approx 0.24$; Spambase, $\pi \approx 0.39$; Phishing, $\pi \approx 0.44$), we compute the universal bracket slack w^* and the marginal-aware slack $w^*(\pi_*)$ for the partitions of E1 and E2, and report the slack reduction. *Status: pending — spec in `experiments/PLANS.md`.*

E3 — MPNN/WL bracket on real graphs. This experiment turns Theorem 16 into a quantitative diagnostic on five canonical node-classification benchmarks. We recall the architectural picture. Xu et al. (2019) prove that every aggregation-based GNN is, in distinguishing power, *at most as fine as the* 1*-WL colouring* (their Lemma 2), and that this ceiling is *achievable* whenever the neighbour aggregator and the readout are both injective on multisets (their Theorem 3, instantiated by the GIN update $(1+\epsilon^{(k)})h_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)}$ with irrational ϵ). Morris et al. (2019) state the same dichotomy in their notation: the 1-WL refinement

Dataset	π_*	w^*	$w^*(\pi_*)$
UCI Adult	TBD	0.1610	TBD
Spambase	TBD	0.1610	TBD
Phishing	TBD	0.1610	TBD

Table 4: E2b: marginal-aware slack reduction (Theorem 4) on real unbalanced labels.

always refines the colouring produced by any 1-GNN at the same depth (their Theorem 1), and there exist weight matrices realising equality whenever distinct initial labels are encoded by linearly independent vectors (their Theorem 2). For depth L, the 1-WL partition Π_L is therefore the *finest cell structure* attainable by *any* depth-L MPNN. Plugging Π_L into our sandwich (2) therefore brackets the empirical training error of *every* admissible depth-L MPNN: the upper endpoint $\frac{1}{2}H(f \mid \Pi_L)$ is realised training-free by a per-cell majority head, and the lower endpoint $H_{\text{bin}}^{-1}(H(f \mid \Pi_L))$ is the Fano-type floor that no architecture in this class can break. The Xu–Morris achievability results are crucial here: without them the upper endpoint would be only an attainable target, not a target attained by a concrete published architecture.

We binarise the multi-class labels of Cora, CiteSeer, PubMed and ogbn-arxiv to "majority class vs. rest"; Twitch-EN ships a native binary mature-content flag. The initial colour $h_v^{(0)}$ is the vertex degree (consistent with Theorem 1 of Morris et al. (2019)); each refinement round is one pass of the 1-WL multiset hash, implemented with sparse-CSR adjacency. *No MPNN is ever instantiated; no GPU is used.* On a single laptop core, WL refinement to $L = 3$ on ogbn-arxiv (169,343 vertices, $\sim$ 1.16M undirected edges) takes 2.7 seconds, and the bracket evaluation is a single $O(|V|)$ sweep on top.

Graph	$\lvert V\rvert$	L	m_L	lower	$\varepsilon^*_{\Pi_L}$	upper
Twitch-EN	7,126	0	130	0.3853	0.4164	0.4809
Twitch-EN	7,126	3	6,648	0.0073	0.0267	0.0312
Cora	2,708	3	2,363	0.0076	0.0292	0.0323
CiteSeer	3,327	3	2,044	0.0400	0.0775	0.1212
PubMed	19,717	3	12,990	0.0205	0.0511	0.0722
ogbn-arxiv	169,343	3	161,943	0.0005	0.0021	0.0029

Table 5: E3: empirical training-label bracket of the 1-WL partition Π_L at depth L on five real graphs (CPU only, no MPNN instantiated). The upper endpoint $H(f \mid \Pi_L)/2$ is the training error of the per-cell majority-vote head; by Xu et al. (2019, Theorem 3) and Morris et al. (2019, Theorem 2) it is realised by a sufficiently wide injective MPNN. The lower endpoint $H_{\text{bin}}^{-1}(H(f \mid \Pi_L))$ is the Fano floor that no depth-L MPNN can break, by Xu et al. (2019, Lemma 2) / Morris et al. (2019, Theorem 1) combined with our (2). Full funnels $L = 0, \ldots, L_{\max}$ in `experiments/results/e3.json`; per-dataset funnel figures in `experiments/figures/e3_*_funnel.pdf`.

What the numbers say. Two regimes are visible in Table 5. (i) At $L = 0$ (degree-binned colouring, no message passing yet) the bracket has substantial width: on ogbn-arxiv the width is 0.151, within 1% of the universal ceiling $w^* \approx 0.161$ of Theorem 3; on Twitch-EN it is 0.096. The achievable error $\varepsilon^*_{\Pi_0}$ sits neither at the floor nor at the ceiling but interior to the bracket — the position within the bracket is a *diagnostic of the partition* (high inside the bracket = decisive but noisy cells; low inside the bracket = pure but small cells). To our knowledge no prior MPNN-expressivity study reports this signal. (ii) At $L = 3$ the bracket has collapsed by an order of magnitude across every dataset; on ogbn-arxiv the cell count $m_3 = 161{,}943$ reaches 95.6% of $|V|$, on Cora 87%, on Twitch-EN 93%. This is consistent with the well-known phenomenon that 1-WL stabilises after a small number of rounds on sparse real graphs (Morris et al., 2019; Arvind et al., 2015); it is also a structural form of overfitting: by depth 3 the 1-WL fingerprint has separated almost every vertex from every other, so the bracket pinches towards zero not because some MPNN secretly solved the task but because the partition has memorised the structure of the graph itself. We report both regimes because the $L = 0$ rows are where the bracket actively bounds; the $L = 3$ rows are the structural ceiling of the 1-WL family.

What this experiment establishes, precisely. First, the bracket is well-defined and tight enough to be checkable on real graphs: every funnel passes the audit gates sub-partition, monotonicity of H and ε^* in L (Theorem 9), the sandwich (2), and the width bound $\frac{1}{2}H - H_{\text{bin}}^{-1}(H) \le w^*$ (Theorem 3), with zero violations across all $\sum_d(L_{\max}(d)+1)$ rows. Second, the bracket quantifies what was previously a qualitative claim. The Xu and Morris papers say "no MPNN exceeds 1-WL"; we say "no depth-L MPNN achieves training error below $H_{\text{bin}}^{-1}(H(f \mid \Pi_L))$, and the witness training-free classifier already achieves $\frac{1}{2}H(f \mid \Pi_L)$." Third, the experiment exposes a property of the benchmarks themselves: ogbn-arxiv is structurally near-trivial for 1-WL (the 1-hop + degree fingerprint already separates 160,000 vertices); CiteSeer is the hardest of the canonical trio, with $m_3/|V| = 0.61$ and $\varepsilon^*_{\Pi_3} = 0.078$; Twitch-EN's mature-content flag is nearly a function of 3-hop neighbourhoods (a 15× collapse in ε^* from $L=0$ to $L=3$).

What this experiment does not establish. The bracket is computed on the full transductive label vector and is therefore a statement about *fitting* power, not generalisation; the population analogue is Theorem 19, empirically exercised on tabular data in E7. The achievability theorem of Xu et al. (2019) requires unbounded MLP width and an injective sum aggregator on a countable feature space; in practice (finite width, GCN/GAT aggregators that are *not* injective on multisets) a trained MPNN sits strictly above $H_{\text{bin}}^{-1}(H(f \mid \Pi_L))$, and the gap to the lower bound is itself an interesting gauge of how much of the 1-WL ceiling a given architecture realises. Finally, the bracket inherits the headline limitation of the 1-WL/MPNN universe: structures indistinguishable by 1-WL (e.g. the triangle-vs.-4-cycle example of Morris et al. (2019, Sec. 4.1)) collapse into the same cell and cannot be separated by either side of the sandwich. Lifting the bracket to the k-WL partitions of Morris et al. (2019, Sec. 5) is immediate (our framework is colouring-agnostic) and is the natural next experiment.

E4 — Cross-architecture duel at fixed cell budget (UCI Adult, $K = 16$). Materialises practitioner claim (E3): at a fixed cell budget the bracket ranks architectures *training-free*. *Status: completed; rebuts the “w^* slack is too wide to be decision-useful” critique by exhibiting two architectures whose brackets order strictly despite each having width $\approx w^*$.*

Architecture	m	H	lower	ε^*_Π	upper	fit (s)
Decision tree (16 leaves)	16	0.4823	0.0992	0.1511	0.2411	0.18
k-means ($k = 16$)	16	0.6597	0.1709	0.2352	0.3299	4.36

Table 6: E4: the supervised tree dominates the unsupervised quantiser at equal cell budget. Figure: `experiments/figures/e4_duel_table.pdf`.

E5 — Achievable-region scatter (synthetic). Visualises Theorem 2 and Theorem 3: 10^3 random partitions of $V = \{1, \ldots, n\}$, $n \in \{4, \ldots, 32\}$, plotted in the (H, ε) plane against the Fano and Hellman–Raviv boundaries. Empirical max upper slack is 0.16096404744368120 vs analytic 0.16096404744368115 (≤ 1 ulp of `Float64`). *Status: completed.* Figure: `experiments/figures/e5_achievable_region_scatter.pdf`.

E6 — Cost comparison vs one training epoch (planned). Quantifies the $O(|V|)$ remark of § 1. On each dataset of the benchmark menu, we time $T_{\text{bracket}} :=$ one call to `bracket_from_cells` against $T_{\text{epoch}} :=$ one CART fit, one k-means `fit_predict`, and one logistic-regression epoch on one-hot cell membership. The ratio $T_{\text{bracket}}/T_{\text{epoch}}$ should be $\ll 10^{-2}$ across the menu. *Status: pending — spec in `experiments/PLANS.md`.*

Dataset	$\lvert V\rvert$	T_{bracket} (ms)	T_{epoch} (ms)	ratio
Spambase	4,601	TBD	TBD	TBD
MAGIC	19,020	TBD	TBD	TBD
UCI Adult	45,222	TBD	TBD	TBD
MNIST-bin	70,000	TBD	TBD	TBD

Table 7: E6: cost of evaluating the bracket vs one training epoch (planned).

E7 — Real-data Proposition 7 concentration (planned). Empirical companion to Theorem 19. On UCI Adult, draw subsamples of size $n \in \{500, 1000, 5000, 10000, 20000, 45000\}$; for each n, $K = 200$ trials, compare the empirical bracket to the full-data bracket via the gap $\Delta_n = |\varepsilon^*_{\text{full}} - \varepsilon^*_{\text{sub}}| + |H_{\text{full}} - H_{\text{sub}}|$ and check empirical coverage at the Hoeffding rate $\kappa(\delta, \eta)\sqrt{\log(4m/\delta_{\text{conf}})/n}$. *Status: pending — spec in* `experiments/PLANS.md`.

n	$\overline{\Delta_n}$	$\Delta_n^{(p95)}$	analytic bound	coverage
500	TBD	TBD	TBD	TBD
5000	TBD	TBD	TBD	TBD
45000	TBD	TBD	TBD	TBD

Table 8: E7: real-data concentration of the empirical bracket to the population bracket as n grows (planned).

10 Related work

Entropy bounds Bayes error. The upper inequality (5) is Hellman & Raviv (1970); closely related entropy upper bounds on Bayes error appear in Tebbe & Dwyer III (1968) and Kovalevskij (1968). The lower inequality is Fano's (Fano, 1961) in the sharp binary form of Cover & Thomas (2006, Thm. 2.10.1). The achievable-region packaging is due to Feder & Merhav (1994), and the most complete modern treatment of two-sided entropy–error inequalities, including extensions and continuous-alphabet generalisations, is Ho & Verdú (2010). Sason & Verdú (2018) place these inequalities in the broader Arimoto–Rényi framework, and Santhi & Vardy (2006) sharpen the lower side under a different optimisation. Hashlamoun et al. (1994) strengthen the upper bound for $M > 2$ classes via f-divergences. The prior-aware sharpening of the Fano side, which we do not pursue here, originates with Han & Verdú (1994, eq. 6).

Partition-constant predictors. The decision-tree literature (Breiman et al., 1984) treats leaf-conditional entropy as an empirical splitting criterion; the explicit *bracket* interpretation as a sharp two-sided bound on the achievable error of any leaf-constant predictor does not appear to have been recorded. Vector quantisation (Gersho & Gray, 1991; Jégou et al., 2011) and product quantisation share the same partition structure; the bracket is, to our knowledge, similarly new in that literature.

MPNN expressivity. The Weisfeiler–Leman ceiling on MPNN expressivity is the central structural result of the last decade (Xu et al., 2019; Morris et al., 2019); refinements include fine-grained expressivity bounds (Böker et al., 2023) and higher-order variants (Morris et al., 2020). Theorem 17 recasts the WL ceiling as one half of an information-theoretic bracket, giving the matching upper bound (constructive plug-in predictor on WL cells) on equal footing.

11 Discussion and limitations

The bracket of Theorem 2 is uniform in Π and f, closed-form, and elementary to evaluate; the uniform slack $w^* \approx 0.1610$ of Theorem 3 pins the irreducible ambiguity that any closed-form bracket of this shape must carry, by Theorem 10. Theorem 4 sharpens the upper side when the label marginal is unbalanced.

From empirical to population. The bracket as stated is an *empirical* statement on a finite V. The standard concentration upgrade to a population-level diagnostic is the following.

Proposition 19 (Population extension). *Let $\mathcal{X}$ be a measurable space, μ a probability distribution on $\mathcal{X}$, and Π a measurable partition of $\mathcal{X}$ into m cells with population masses $\mu(C) \geq \delta > 0$ and population means $\bar{P}_C := \mathbb{E}[f \mid C] \in [\eta, 1-\eta]$ for some $\eta > 0$. Let $V = (v_1, \ldots, v_n)$ be i.i.d. draws from μ and write ε^*_Π, $H(f \mid \Pi)$ for the empirical quantities of § 2 on V. Define the population analogues*

$$\epsilon^*_{\Pi,\mu} := \sum_C \mu(C)\, \min(\bar{P}_C, 1 - \bar{P}_C), \qquad H_\mu(f \mid \Pi) := \sum_C \mu(C)\, H_{\text{bin}}(\bar{P}_C).$$

Then for every $\delta_{\text{conf}} \in (0,1)$, with probability at least $1-\delta_{\text{conf}}$ over the draw of V,

$$\left|\,\epsilon^*_{\Pi,\mu} - \varepsilon^*_{\Pi}\,\right| \;+\; \left|\,H_\mu(f \mid \Pi) - H(f \mid \Pi)\,\right| \;\le\; \kappa(\delta,\eta)\sqrt{\frac{\log(4m/\delta_{\text{conf}})}{n}}, \tag{6}$$

where $\kappa(\delta,\eta) := \frac{1}{\sqrt{\delta}}\bigl(1+\log_2\frac{1-\eta}{\eta}\bigr)$. In particular the population bracket

$$H_{\text{bin}}^{-1}\bigl(H_\mu(f \mid \Pi)\bigr) - \kappa\sqrt{\tfrac{\log(4m/\delta_{\text{conf}})}{n}} \;\le\; \epsilon^*_{\Pi,\mu} \;\le\; \tfrac{1}{2}H_\mu(f \mid \Pi) + \kappa\sqrt{\tfrac{\log(4m/\delta_{\text{conf}})}{n}} \tag{7}$$

holds with the same probability.

Proof sketch. Hoeffding's inequality applied to the indicator $\mathbf{1}\{v \in C\}$ gives $|q_C - \mu(C)| \le \sqrt{\log(4m/\delta_{\text{conf}})/(2n)}$ for each fixed C with probability $\ge 1-\delta_{\text{conf}}/(2m)$; union bound over C. Conditioned on the cell counts, the $|C|$ labels in C are i.i.d. Bernoulli($\bar{P}_C$), so Hoeffding again yields $|P_C - \bar{P}_C| \le \sqrt{\log(4m/\delta_{\text{conf}})/(2n\mu(C))}$ on the same high-probability event. On $[\eta, 1-\eta]$, H_{bin} is Lipschitz with constant $\log_2((1-\eta)/\eta)$ and $p \mapsto \min(p, 1-p)$ is 1-Lipschitz; weighting by $\mu(C)$, summing over cells, and using $\mu(C) \ge \delta$ to bound $\sum_C \sqrt{\mu(C)/n} \le \sqrt{m/n} \le \sqrt{1/(n\delta)}$ absorbs all $\mathcal{O}(\sqrt{\log m/n})$ contributions into the stated constant. See Antos & Kontoyiannis (2001) for a refined analysis without the uniform-mass assumption. □

Theorem 19 converts Theorem 2 into a PAC-style population diagnostic: with high probability, the training-set bracket transfers to the underlying distribution up to an additive $\mathcal{O}\bigl(\sqrt{\log m/n}\bigr)$ correction that vanishes as $n \to \infty$. The boundedness assumptions $\mu(C) \ge \delta$ and $\bar{P}_C \in [\eta, 1-\eta]$ are standard for finite-alphabet plug-in entropy estimation; relaxing them requires the more careful analysis of Antos & Kontoyiannis (2001). A Monte-Carlo verification of the concentration rate is included in the extended verifier `verify_t4_population.py`; see § 9.4.

Limitations. (a) Binary labels only; multi-class via the Feder–Merhav concave envelope is mechanically routine but loses the closed-form simplicity of H_{bin}^{-1}. (b) The bracket bounds the minimum risk *given* the partition; designing partitions that minimise $H(f \mid \Pi)$ for a given budget is the harder problem the bracket does not directly address. (c) The population extension of Theorem 19 requires $\mu(C) \ge \delta$ and $\bar{P}_C \in [\eta, 1-\eta]$; entropy estimation under vanishing cell mass needs the refined treatment of Antos & Kontoyiannis (2001).

Future work. A fully prior-aware tightening of the *lower* side via data processing on the binary error indicator (Han & Verdú, 1994; Ho & Verdú, 2010) can sharpen Theorem 4's bound when the label marginal is bounded away from 1/2; we defer this to a companion paper. A mechanised proof in Lean 4 / mathlib4, leveraging the recent additions of `Real.binEntropy` and the `ConcaveOn` API, would give a kernel-checked counterpart to `verify.jl` and is in preparation.

12 Conclusion

We have given a single two-sided closed-form bracket on the partition-restricted minimum risk of every partition-constant predictor of a binary label, in the form ML practitioners need: elementary proof, a universal slack constant $w^* \approx 0.1610$ (its main theoretical novelty), a marginal-aware refinement that shrinks this slack for unbalanced labels, a population extension under i.i.d. sampling, three worked applications, and an extended verification ladder in Julia, SymPy, and Monte Carlo. The bracket makes the partition the explicit unit of expressivity, lifting classical entropy–error inequalities into an optimisation-free, architecture-level diagnostic for decision trees, vector quantisers, and graph neural networks alike.

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